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Roll No

ME-5001-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Turbo Machinery

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Derive Euler turbine equation.
b) What is the condition for maximum utilization factor for impulse turbine?
2. In a vertical-shaft Francis turbine the available head at the inlet flange of the turbine is 150m and the vertical distance between the runner and the failrace is 1.95m. The runner tip speed is 34.5m/s. the velocity of the water entering the runner is 39m/s and the velocity of the water exiting the runner is constant and equal to 10.5m/s. The flow velocity at the exit of the draft tube is 3.45m/s. The hydraulic energy losses estimated from the turbine are equal to 6m at the volute 1.05m at the draft tube, and 9.9m at the runner. Determine the pressure head (with respect to the failrace) at the inlet and exit of the runner, the flow angle at the runner, inlet and the efficiency of the turbine.
3. In a De-laval turbine, the steam enters the wheel through a nozzle with a velocity of 500m/s and at an angle of 20° to the direction of motion of the blade. The blade speed is 200m/s and the exit angle of the moving blade is 25° . Find the inlet angle of the moving blade, exit velocity of steam and its direction and work done per kg of steam.

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4.
 - a) What are the advantages of centrifugal pump over reciprocating pump. Explain in brief.
 - b) Classify hydraulic turbines.
 - c) Explain the phenomenon of cavitation in water turbines.

5. A Pelton wheel is to be designed for the following specifications:
Shaft power = 11,772 kW; Head 380 metres; speed = 750 r.p.m.; overall efficiency = 86%; jet diameter is not to exceed one - sixth of the wheel diameter. Determine:
 - i) The wheel diameter
 - ii) The number of jet required
 - iii) Diameter of the jet.Take $K_{v_1} = 0.985$ and $K_{u_1} = 0.45$

6.
 - a) Compare the axial flow and centrifugal compressors.
 - b) Define surging and efficiency of Axial flow compressor.

7.
 - a) What is reheat factor? What causes the improvement in efficiency of multi stage turbine as compared with a single stage turbine with same stage efficiency?
 - b) Why is Kaplan turbine efficiency nearly constant, irrespective of speed variation under varying load?

8. Write short notes on:
 - a) Dimensional analysis
 - b) Polytropic efficiency of compressor
 - c) Axial flow compressor
 - d) Draft lube
